

CONCEPTUAL PROJECT-1

AI SMART SAFETY HELMET

BUILDING THE FUTURE



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INTRODUCTION



- Construction accidents are common.
 - Existing helmets offer limited protection.
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- Need for smart wearable safety.
 - AI and IoT can improve worker safety.
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PROBLEM STATEMENT

1. Delayed Rescue

- After an accident, workers may not be able to call for help.
- Supervisors often learn about the accident too late.

2. Lack of Continuous Monitoring

- Traditional safety helmets do not monitor the worker's health or surroundings.
- Supervisors cannot track the worker's condition or location in real time.

3. No Accident Prediction

- Existing safety systems mainly respond after an accident happens.
- They cannot identify warning signs or predict hazardous situations in advance.
- As a result, preventive action is often not possible.

4. Worker Safety Challenges

- Construction workers face multiple risks every day, such as:
 - Falling objects
 - Slips and falls from heights
 - Exposure to toxic gases





FEATURES

Sensors:

- Impact Detection
- Fall Detection
- GPS Tracking
- Emergency SMS

CORE V1 FEATURES

- Impact Detection
- Fall Detection
- LoRa Communication
- Underground Worker Status
- Gas / Smoke Detection
- Temperature Monitoring

METHODOLOGY

Explain the working flow:

Worker wears helmet



Sensors collect data



ESP32 processes data



AI checks risk level



Emergency alert sent



Supervisor Dashboard updates



PROPOSED INNOVATIVE FEATURES



Point 01

AI Hazard Memory

AI remembers:

- Near misses
- Unsafe locations
- Previous accidents

Predicts dangerous zones.



Point 02

Autonomous Team Intelligence

Multiple helmets share sensor data.

If many helmets detect abnormal conditions, AI predicts danger before an accident.



Point 03

Smart Rescue Prioritization

When multiple accidents happen, AI ranks workers by severity using:

- Heart Rate
- Impact Force
- Movement
- Location

This helps rescue teams respond faster.

ADVANTAGES



01

Faster Emergency Response

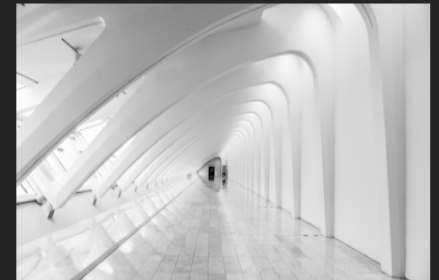
- Accidents are detected instantly.
- Emergency alerts are sent automatically.
- This helps medical teams reach workers faster.



02

Improved Worker Safety

- The helmet continuously monitors the worker.
- It detects dangerous situations in real time.
- This helps prevent serious accidents.



03

Reduced Fatalities

- Quick accident detection enables faster medical assistance.
- Early response reduces the severity of injuries.
- This can help save lives.

APPLICATIONS & FUTURE SCOPE

Applications

- Construction sites
- Mining
- Oil & Gas Industry
- Factories
- Tunnel construction
- Disaster rescue teams

Future Scope

- AI accident prediction
 - Digital twin of construction site
 - Integration with drones
 - Cloud dashboard
 - Voice assistant
 - AR safety guidance
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EXPECTED OUTCOMES

Point 01

Faster Emergency Alerts

- The system detects accidents instantly.
- Emergency alerts are sent automatically to supervisors and the medical team.

Point 02

Reduction in Accident Response Time

- Automatic alerts eliminate delays in reporting accidents.
- This allows medical assistance to reach the worker faster.



PROJECT FOCUS

- Designed for underground workers where cellular communication may be unavailable.
- LoRa is used as the primary short-message wireless link between helmets and a gateway.
- The helmet focuses on impact/fall detection, environmental monitoring and emergency communication.
- Surface communication can be handled by the gateway rather than by each helmet.

UNDERGROUND

LOW POWER

LoRa

SURFACE ALERT

WHY LoRa FOR UNDERGROUND WORK

- Underground workers may operate beyond reliable cellular coverage.
- LoRa supports low-power transmission of small safety messages.
- Multiple helmets can communicate with a common gateway.
- Gateway placement and antenna position must be tested for the actual tunnel environment.
- Communication range underground must be validated experimentally; no fixed range is assumed.

NO CELL

LOW POWER

MULTI-NODE

TEST RANGE

SYSTEM ARCHITECTURE

- Helmet Sensors → ESP32-S3 → LoRa → Underground Gateway → Surface Station → Supervisor
- MPU6050 detects impact, abnormal motion and fall patterns.
- MQ-2 provides gas/smoke indication; DHT22 monitors temperature/humidity.
- LoRa carries worker ID, alert status and selected sensor values.
- The gateway forwards critical events to the surface monitoring system.

SENSORS

ESP32-S3

LoRa

GATEWAY

SURFACE

ESSENTIAL HARDWARE COMPONENTS

- ESP32-S3 — main controller
- LoRa transceiver — primary underground communication
- MPU6050 — impact + fall detection
- MQ-2 — gas/smoke indication
- DHT22 — temperature/humidity
- Li-ion/Li-Po battery — helmet power
- TP4056 charging module + 5 V boost converter
- LoRa gateway/receiver + antenna
- Helmet enclosure, perfboard, wires, switch and LEDs

CONTROL

IMPACT/FALL

GAS/TEMP

LoRa

UNDERGROUND HELMET WORKFLOW

- 1. Sensors continuously monitor the worker and environment.
- 2. ESP32-S3 analyzes the sensor readings.
- 3. A risk state is created: NORMAL, WARNING or CRITICAL.
- 4. Critical events are transmitted through LoRa.
- 5. Gateway receives the alert and forwards it to the surface.
- 6. Supervisor identifies the affected helmet/worker and tunnel zone.

SENSE

ANALYZE

LoRa

GATEWAY

ALERT

ACCIDENT DETECTION LOGIC

- Sudden acceleration / impact is detected using MPU6050.
- Orientation and movement are checked after the event.
- Impact + abnormal orientation + inactivity → CRITICAL.
- Gas/smoke or temperature threshold crossing → WARNING.
- CRITICAL status → immediate LoRa emergency packet.
- System can reduce false alarms by checking multiple conditions before escalation.

IMPACT

ORIENTATION

INACTIVITY

CRITICAL

LoRa EMERGENCY PACKET

- Each helmet sends a compact data packet instead of large data streams.
- Example: Helmet ID | Risk Level | Impact | Gas Status | Temperature | Battery | Zone/Node
- Normal packets can be sent periodically.
- Critical packets should be transmitted immediately and repeated until acknowledged.
- Gateway acknowledges received critical packets where the selected LoRa system supports it.

ID

RISK

SENSORS

BATTERY

ZONE

MULTI-HELMET TEAM INTELLIGENCE

- Several helmets can share safety status through the LoRa network.
- One critical event can be visible to the gateway and surface control room.
- The system can prioritize alerts using impact severity, movement and worker zone.
- Future versions can add multiple gateways/repeaters for larger underground areas.

HELMET 01

HELMET 02

GATEWAY

CONTROL

PROTOTYPE TESTING

- Normal movement → no emergency alert.
- Controlled impact simulation → impact packet received by gateway.
- Simulated fall + inactivity → CRITICAL LoRa alert.
- Gas/smoke test → WARNING packet.
- Temperature test → sensor value transmitted.
- Multiple helmets → verify that each helmet ID is correctly received.
- Tunnel test → measure actual LoRa reliability at different positions.

IMPACT

FALL

GAS

MULTI-NODE

TUNNEL

LIMITATIONS & FUTURE SCOPE

- LoRa performance underground depends on tunnel geometry, rock/soil, antenna placement and network configuration.
- MQ-2 is a prototype gas/smoke indication sensor, not a complete industrial toxic-gas monitor.
- GPS/GNSS is not the primary underground positioning method in this version.
- Future: LoRa repeaters/gateways for deeper tunnels.
- Future: underground zone/anchor-based localization.
- Future: camera-based PPE and hazard recognition.
- Future: cloud dashboard, ML prediction, digital twin and drone-assisted rescue.

REPEATERS

LOCALIZATION

CAMERA AI

ML

CLOUD

FINAL PROTOTYPE CONCLUSION

- The redesigned prototype is specifically focused on underground workers.
- LoRa replaces cellular communication as the helmet's primary underground wireless link.
- Core chain: Sense → Detect → LoRa → Gateway → Surface Alert.
- The design is simpler, lower-power and more realistic to prototype.
- Advanced camera AI and other features remain future upgrades.

SENSE

DETECT

LoRa

GATEWAY

ALERT



THANK YOU

"Our project combines IoT and AI to improve construction worker safety. Instead of only detecting accidents, our proposed innovations aim to predict risks, coordinate intelligent teamwork between helmets, and prioritize rescue efforts. This can help create a safer and smarter construction environment."

"Prevention is better than rescue, and intelligence is better than reaction."

Thank You